Deductive reasoning

1 Write the definition for when an argument is:

a Valid

b Invalid

2 What is a fallacy?

3 Identify the form of argument that each of the following represents:

a
$$\begin{array}{l} a \implies b \\ \sim a \\ \hline \therefore \sim b \end{array}$$

b
$$\begin{array}{l} j \implies k \\ j \\ \hline \therefore k \end{array}$$

c
$$\begin{array}{l} j \vee k \\ \sim j \\ \hline \therefore k \end{array}$$

d
$$\begin{array}{l} j \implies k \\ k \\ \hline \therefore j \end{array}$$

e
$$\begin{array}{l} j \implies k \\ k \implies l \\ \hline \therefore j \implies l \end{array}$$

f
$$\begin{array}{l} j \implies k \\ \sim k \\ \hline \therefore \sim j \end{array}$$

4 State whether each of the following arguments is valid or invalid.

a
$$\begin{array}{l} n \implies o \\ \sim n \\ \hline \therefore \sim o \end{array}$$

b
$$\begin{array}{l} a \vee b \\ \sim a \\ \hline \therefore b \end{array}$$

c
$$\begin{array}{l} x \implies y \\ x \\ \hline \therefore y \end{array}$$

d
$$\begin{array}{l} a \implies b \\ b \implies c \\ \hline \therefore a \implies c \end{array}$$

e
$$\begin{array}{l} a \vee b \\ \sim b \\ \hline \therefore a \end{array}$$

f
$$\begin{array}{l} t \implies u \\ \sim u \\ \hline \therefore \sim t \end{array}$$

g
$$\begin{array}{l} e \implies f \\ f \\ \hline \therefore e \end{array}$$

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5 Consider each of the following arguments.

i Write the argumet in symbolic form.

ii Determine whether the argument is valid or invalid.

a If you log on to Facebook during class, then the teacher gets angry.
You do not log on to Facebook during class.

$\therefore$The teacher does not get angry.

Let
e: You log on to Facebook during class.
f: The teacher gets angry.

b If the phone is made by SuperFone, then the phone is high quality.
The phone is high quality.

$\therefore$The phone is made by SuperFone.

Let
l: The phone is made by SuperFone.
m: The phone is high quality.

c If we go to the park, then we see a dog.
We go to the park.

$\therefore$ We see a dog.

Let
j: We go to the park.
k: We see a dog.

d If I catch the 9 : 30am bus, then I am on time for the start of work at10 : 00am
If I am on time for the start of work at 10 : 00 am, then I can make it to the party at6 : 00 pm.

$\therefore$ If I catch the 9 : 30 am bus, then I can make it to the party at6 : 00 pm.

Let
a: I catch the 9 : 30 am bus.
b: I am on time for the start of work at 10 : 00 am.
c: I can make it to the party at 6 : 00 pm.

e If the car is a Falcon, then the car is made by Ford.
The car is not made by Ford.

$\therefore$The car is not a Falcon.

Let
a: The car is a Falcon.
b: The car is made by Ford.

f If we travel by car, then we see kangaroos.
We see kangaroos.

$\therefore$We travel by car.

Let
c: We travel by car.

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g If you study education, then you can become a teacher.
You study education.

$\therefore$ You can become a teacher.

Let
j: You study education.
k: You can become a teacher.

h If you find a cure for motor neuron disease, then you are a hero.
You are not a hero.

$\therefore$ You did not find a cure for motor neuron disease.

Let
r: You find a cure for motor neuron disease.
s: You are a hero.

i If it is windy, then the ceremony will be held indoors.
If the ceremony is held indoors, then the after-party will be cancelled.

$\therefore$ If it is windy, then the after-party will be cancelled.

Let
a: It is windy.
b: The ceremony will be held indoors.
c: The after-party will be cancelled.

j If Tom is the midfielder in the team, then Iain is the goalkeeper.
Tom is not the midfielder in the team.

$\therefore$ Iain is not the goalkeeper.

Let
a: Tom is the midfielder in the team.
b: Iain is the goalkeeper.

6 Consider each of the following statements.

i Write the argumet in symbolic form.

ii Determine whether the argument is valid or invalid.

a " If you study medicine then you understand biology. You do not understand biology. Therefore, you did not study medicine."
Let
a: You study medicine.
b: You can understand biology.

b " The car has a blown engine or the car does not have charged battery. The car has charged battery. Therefore, the car has a blown engine."
Let
l: The car has a blown engine.
m: The car has charged battery.

c "Charlie is watching TV in the bedroom with the lights off or Rochelle is wearing a blindfold. Charlie is not watching TV in the bedroom with the lights off. Therefore, Rochelle is wearing a blindfold."
Let
r: Charlie is watching TV in the bedroom with the lights off.
s: Rochelle is wearing a blindfold.

d "If the cat is here, then the parrot is hiding. The cat is not here. Therefore, the parrot is not hiding."
Let
g: The cat is here.
h: The parrot is hiding.

e "If Georgia graduates from law school, then she will be become a lawyer. Georgia will become a lawyer. Therefore, Georgia graduated from law school."
Let
t: Georgia graduates from law school.
u: Georgia will become a lawyer.

7 Determine whether each of the following is valid or not valid.

a

- If Y, then B.
- Y is true.
- Therefore, B is true.

b

- All nugnugs are blubbits.
- I am a nugnug.
- Therefore, I am a blubbit.

c

- All squares are rhombuses.
- All rhombuses are parallelograms.
- Therefore, all squares are parallelograms.

d

- All swoopitips are blubbits.
- All swoopitips are vostions.
- Therefore, all blubbits are vostions.

e

- If it is nighttime, then the light in my room is on.
- The light in my room is on.
- Therefore, it is nighttime.

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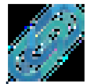
8 Consider the following statements: 
What can be deduced from these statements.

a

- All lemurs have a jaw.
- Holly has a jaw.

b

- All pairs of lines in the plane are either parallel or intersecting.
- $\overleftrightarrow{AB}$ and $\overleftrightarrow{XY}$ are not parallel.

9 State a valid conclusion to complete the argument:

a When a blubbit leaves its nest, a swoopitip always follows. A swoopitip is a big version of a nugnug. A plippi can force a blubbit to leave its nest with its odor.

b Every laborer works from blueprints. All plumbers are laborers. Luigi is a laborer.

10 Consider the following arguments.

- All jibboos are mumpers.
-
- All lobies are snoozles.
- Therefore, all lobies are mumpers.

Write two statements that can be used to fill the second bullet point and make the argument valid.

11 Write a valid deduction for the following group of statements.

- If a yahyok breathes underwater, then it is a plippi.
- If a yahyok has four feet, then it breathes underwater.
- Linus is a yahyok that has four feet.

Additional questions

12 Write the logical argument $p \to q$ as a conditional statement.

13 Use the Law of Detachment to make a conclusion from each pair of premises.

a If a number is a multiple of two, then it is even.
All multiples of four are multiples of two.

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b If a triangle is not isosceles, then it is scalene.
The triangle T is not isosceles.

14 Use the Law of Syllogism to write a new conditional statement from the given statements.

a If a regular polygon has five sides, then it is a pentagon.
If a polygon is a regular pentagon, then it has an interior angle sum of 540°.

b If the sum of the digits of a number is a multiple of 9, then the number is divisible by 9.
If a number is divisible by 9, then it is divisible by 3.

15 Consider the given information:

- All rectangles are parallelograms
- If a quadrilateral is a parallelogram, then it has two pairs of opposite congruent sides.
- If a quadrilateral has four congruent sides, then its a rhombus.

Determine whether the following arguments are valid or not.

a A quadrilateral has four congruent sides. Therefore its a rhombus.

b A quadrilateral has two pairs of opposite congruent sides. Therefore its a rectangle.

c A quadrilateral is a rectangle. Therefore it has two pairs of opposite congruent sides.

16 For each pair of premises, is it possible to deduce a conclusion? If not, explain why.

a If you play basketball, then you play a sport.
Ishaq plays a sport.

b If Cyrille wears socks to sleep, she will wake up early.
Cyrille wears socks to sleep on New Years Eve.

c If Airon is going to Everglades National Park, then it will be for a geography class.
Airon is at Everglades National Park.

17 Consider the given information:

- $p \to q$
- $q \to r$
- $s \to t$

Determine whether the following arguments are valid or not.

a Suppose that p is true. Therefore r is true.

b Suppose that q is not true. Therefore r is not true.

c Suppose that s is true. Therefore t is true.

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18 Uxia presents the following argument:

- All beetles are insects.
- If a bug has six legs then it is an insect.
- A ladybug is a beetle.
- Therefore, a ladybug has six legs.

Is this argument valid? If not, explain why.

19 Levi presents the following argument:

- A quadrilateral is a parallelogram if and only if it's diagonals bisect each other.
- If a parallelogram has at least one right angle, then it is a rectangle.
- If all the sides of a rectangle are congruent, then it is a square.
- The quadrilateral Q has diagonals which bisect each other, at least one right angle and four sides of equal length.
- Therefore, Q is a square.

Explain why Levi's argument is valid.

20 Violette tells her family the following:

- She will be meeting a friend at the park at noon.
- She is going to leave home one hour before noon.
- She will be going directly to the park.

Violette's family deduces that it will take Violette one hour to get from her home to the park.

a Explain why this deduction could be invalid.

b Construct an additional premise to make the family's deduction valid.

21 Construct a set of premises which would result in the deduction:
Puerto plays basketball.

22 Consider the argument:

- All birds are blue.
- All blue objects are chairs.
- A raven is a bird.
- Therefore, a raven is a chair.

a Is this argument valid? Explain.

b Is this argument true? Explain.

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